Chapter 5: Operations

This chapter covers all modules that support day-to-day field and equipment management: recording field operations, managing fuel and equipment, tracking inventory of agrochemicals, logging day labor, scheduling workers, and maintaining service contracts. Read this chapter if you are a Supervisor, Farm Manager, or Accountant responsible for operational data entry.

Who can access Operations?

Supervisors have full access to Operations, Fuel, Maintenance, Cronograma, and Rainfall modules.

Accountants share full operational access and can additionally create transactions linked to operations.

Management and Admin roles have full access to all modules.

Viewers can read all operational data but cannot create or edit records.

5.1 Operations Module Map

The Operations section of Ledger DR is accessible from the left-hand navigation menu. It contains the following sub-modules:

Sub-module	Purpose
Field Operations	Log tractor and manual operations by field, date, implement, and driver
Fuel	Record fuel dispenses and tank refills; monitor equipment fuel consumption
Equipment Maintenance	Track oil-change and routine maintenance events by hour meter
Inventory	Manage agrochemical and input stock; record purchases and usage
Day Labor	Log and close weekly jornalero (day worker) entries by field and task
Cronograma	Weekly work schedule for permanent employees and day workers
Service Contracts	Track third-party equipment contracts by work unit and payment
Rainfall	Record daily rainfall readings across the four farm weather stations

5.2 Farms and Fields

All operations are assigned to a specific field. Fields belong to farms. Understanding the farm-field hierarchy is essential before recording any operation.

5.2.1 Farm List

The system currently manages eight farms:

- La Java
- Madrigal
- Palmarito
- Bebe
- La Virgencita
- Solar
- Caoba
- Carriles

Farms are managed by administrators. If a farm is missing, contact your system administrator — do not create duplicate entries.

5.2.2 Fields

There are 84 fields registered across all farms. Each field has a short code (e.g., C02, BN5, AP01) and a size in hectares. When recording an operation, you select the field from a dropdown. Inactive fields (marked is_active = false) do not appear in the dropdown.

Field codes

Field codes are short alphanumeric identifiers assigned when a field is created. If you cannot find a field in the dropdown, it may be deactivated. Ask your administrator to reactivate it, or verify you have selected the correct farm filter first.

5.3 Recording Field Operations

A field operation is any mechanical or manual agricultural activity performed on a specific field on a specific date — harrowing, planting, spraying, harvesting, and so on. Operations are the core productivity record of the system.

Navigate to Operations > Field Operations and click + New Operation.

5.3.1 Operation Types

The system has 41 active operation types, divided into mechanical (requires a tractor) and manual (no tractor). The type selected determines which other fields are required:

Mechanical vs. manual operations

Mechanical operations require you to select a Tractor and may require an Implement. The Tractor field is optional in the form but strongly recommended — without it, hour meter tracking and fuel reconciliation are impossible.

Manual operations (e.g., Chapeo manual, Siembra manual, Deshierbo manual) do not require a tractor. Enter Workers Count instead.

5.3.2 Operation Form Fields

Field	Required?	Description
Operation Date	Required	The date the work was performed. Defaults to today. Cannot be a future date.
Operation Type	Required	Select from 41 active types. The type determines whether tractor and implement fields are shown.
Field	Required	The specific farm field where the operation took place. Select from the active fields dropdown.
Tractor	Recommended	Required for all mechanical operations. Select from the 10 registered tractors. Leaving this blank disables hour meter tracking and fuel efficiency reporting.
Implement	Optional	The implement attached to the tractor (e.g., Rastra Tatú, Cultivadora, Pulverizador). Not required for transport or simple operations.
Driver	Recommended	Name of the tractor operator. Free text. Strongly recommended — used in productivity reports and equipment accountability.
Start Hours	Recommended	Hour meter reading at the start of the operation. Required for fuel-per-hour and productivity calculations.
End Hours	Recommended	Hour meter reading at the end of the operation. Must be greater than or equal to Start Hours.
Hectares Done	Recommended	Area covered by this operation in hectares. Used to calculate cost-per-hectare and application rates. Defaults to 0 if left blank.
Workers Count	Conditional	Number of manual workers involved. Required for manual operations; optional for mechanical operations.
Notes	Optional	Free-text notes about conditions, issues, or observations relevant to this operation.

Always enter hectares done

Hectares Done is the most important productivity metric. Currently 82% of operations in the system have this field left at zero, making cost-per-hectare reports unreliable. Please enter the actual area completed for every operation, even an estimate.

5.3.3 Tractor and Equipment Reference

The following tractors are registered in the system:

Name	Type / Brand	Current Hours
Pala Volvo	Volvo L70H (loader)	541 h
Shaktiman	Shaktiman 3636 Tejas	51 h
Landini 90S	Landini 90S	2,027 h
JD3006	John Deere 6150R	7,141 h
JD2326	John Deere 6150R	7,190 h
JD 7280R	John Deere 7280R	5,074 h
MF4297	Massey Ferguson 4297	4,660 h
JD 7230R	John Deere 7230R	5,522 h
Drone	Drone (aerial sprayer)	0 h
Contrato	Third-party contract equipment	2,588 h

Available implements include: Cultivadora (Baldan), Rastra Tatú, Rastra Maschio, Mulcher (FAE), Cosechadora (Colhicana), Canterizador (Agromatão), Chapeadora, Pulverizador, Tapador, Sanjeador, Autovolteo, Cobra (Elho), Trimmer (Stihl), and Carreta de cana.

5.3.4 Attaching Inventory Inputs

If the operation involved applying agrochemicals, fertilizers, or other inventory items (e.g., a spraying operation), you can attach the inputs consumed directly to the operation record. This links the field application to inventory stock deductions.

1. Save the operation first.
2. Open the saved operation and scroll to the Inputs section at the bottom of the detail view.
3. Click + Add Input.
4. Select the inventory item from the dropdown (only active items are shown).
5. Enter the quantity used in the item's base unit (liters, kg, or gallons as appropriate).
6. Click Save Input. Repeat for each product applied.

Inputs and stock

Every input you attach records the quantity used against that inventory item. This is the primary mechanism for tracking agrochemical consumption per field and operation. See Section 5.6 for Inventory management.

5.3.5 Editing and Deleting Operations

Operations can be edited by Supervisors, Accountants, and Management at any time. There is no lock on past operations. Use the edit (pencil) icon from the operations list or from the operation detail view.

Deleting an operation also deletes its inputs

If you delete an operation that has attached inputs, all input records for that operation are permanently removed. This will affect inventory consumption reports. Prefer editing over deletion whenever possible.

5.4 Fuel Management

The Fuel module tracks diesel consumption across the farm's three tanks and ten pieces of equipment. There are two types of fuel transactions: refills (diesel added to a tank) and dispenses (diesel dispensed from a tank to a piece of equipment).

5.4.1 Fuel Tanks

Three tanks are currently registered:

Tank Name	Use Type	Capacity	Current Level
Mobile	Agriculture	560 gal	462 gal
Main Tank	Agriculture	1,200 gal	0 gal
Planta Tank	Industry	1,000 gal	110 gal

Tank levels shown may not be current

A known system limitation: fuel dispense transactions currently update the pump counter (last_pump_end_reading) but may not automatically deduct gallons from the displayed tank level. Always verify the physical tank gauge when planning fuel purchases. A fix for automated tank-level deduction is scheduled.

5.4.2 Recording a Fuel Dispense

A dispense is recorded when diesel is pumped from a tank to a tractor or piece of equipment. Navigate to Operations > Fuel > + New Fuel Transaction and select Dispense.

Field	Required?	Description
Tank	Required	Which tank the fuel is being drawn from (Mobile, Main Tank, or Planta Tank).
Equipment	Required	The tractor or equipment receiving the fuel. Select from active equipment list.
Date & Time	Required	The date and time of the dispense. Defaults to now.
Pump Start Reading	Required	The pump odometer reading before dispensing begins. Must be equal to or greater than the previous recorded end reading.
Pump End Reading	Required	The pump odometer reading after dispensing ends. Must be greater than Pump Start Reading. The difference must match the Gallons field.
Hour Meter Reading	Recommended	The equipment's current hour meter at time of fueling. Used to compute gallons-per-hour efficiency. Do not leave blank.
Notes	Optional	Any relevant observation (unusual consumption, equipment status, etc.).

Driver portal submissions

Drivers with the Driver role can submit fuel dispenses directly from a mobile-friendly portal without logging into the main application. These submissions are reviewed in Operations > Fuel > Pending Submissions before being committed to the ledger.

5.4.3 Recording a Tank Refill

A refill is recorded when a fuel delivery adds diesel to one of the tanks. Navigate to Operations > Fuel > + New Fuel Transaction and select Refill.

Field	Required?	Description
Tank	Required	The tank being refilled.
Date	Required	The delivery date.
Gallons	Required	Gallons delivered. Cannot exceed the tank's remaining capacity.
Notes	Recommended	Record the supplier name and delivery reference number (e.g., 'Purchase from Rojas, invoice #4512').

Refill and the accounting module

Large fuel purchases should also be recorded as a transaction in the Accounting > Transactions module (see Chapter 4) to capture the financial cost. The fuel refill in this module records the physical volume only.

5.4.4 Reviewing Fuel Consumption

The fuel list view shows all dispenses and refills in reverse chronological order. Use the filters at the top to narrow by tank, equipment, or date range. Each dispense shows the calculated gallons-per-hour efficiency where an hour meter reading was provided.

Spotting excessive consumption

A significant spike in gallons-per-hour for a specific tractor may indicate a mechanical problem (injector leak, improper idling) or a data entry error in the hour meter reading. Review any reading that differs by more than 30% from the same equipment's historical average.

5.5 Equipment Maintenance

The Maintenance sub-module tracks routine service events for each piece of equipment. Each tractor has a maintenance interval (default 250 hours) that triggers a warning when the current hour meter approaches the next service threshold.

5.5.1 Maintenance Countdown

The equipment list shows each tractor's hours remaining until the next scheduled service. This is calculated as:

Hours until maintenance = Maintenance interval − (Current hour meter − Last maintenance hour meter reading)

When a tractor reaches 0 hours remaining, the system displays a maintenance warning badge on the equipment card and in the operations form when that tractor is selected.

5.5.2 Logging a Maintenance Event

Navigate to Operations > Equipment > select the tractor > Maintenance tab > + Log Maintenance.

Field	Required?	Description
Maintenance Date	Required	The date the service was performed. Defaults to today.
Maintenance Type	Required	Currently only Routine is available. Additional types (Repair, Inspection, Overhaul) may be added by administrators.
Hour Meter Reading	Required	The tractor's hour meter at the time of service. This resets the maintenance countdown from this value.
Notes	Recommended	Record what work was done: oil grade, filter part numbers, technician name, observed issues. This note is permanent and cannot be edited after saving.

Hour meter corrections

The system only accepts hour meter readings equal to or greater than the last recorded reading. If a maintenance entry was made with an incorrect (too-high) hour meter value, the tractor's maintenance countdown will be reset to the wrong baseline. Contact your administrator to correct this — it cannot be fixed by creating a new maintenance record.

5.6 Inventory

The Inventory module manages stock levels for all agrochemicals, fertilizers, and other farm inputs. The current inventory includes 27 items: 14 post-emergent herbicides, 4 pre-emergent herbicides, 3 fertilizers, 2 adjuvants (adherentes), 2 conditioners, and 2 fuel items tracked automatically from tank levels.

5.6.1 Viewing Stock Levels

Navigate to Operations > Inventory to see all active items with their current quantity, purchase unit, use unit, and recommended dose per hectare where configured.

Minimum stock alerts are not yet configured

None of the 27 inventory items currently have a minimum_stock threshold set, so no low-stock alerts will fire. Supervisors should review stock levels manually before planning spray or fertilization operations. Setting minimum stock levels for critical inputs is a recommended administrative task.

5.6.2 Recording a Purchase

When a shipment of agrochemicals or other inputs is received, record it as an inventory purchase to add stock. Navigate to Operations > Inventory > select the item > + New Purchase.

Field	Required?	Description
Purchase Date	Required	The date the goods were received.
Quantity	Required	The quantity received in the item's purchase unit (e.g., liters, kg, gallons).
Unit Price	Required	Price per purchase unit in DOP. Used to calculate average cost.
Total Price	Required	Auto-calculated as Quantity × Unit Price. You may override this if the invoice total differs (e.g., due to a discount).
Packaging Unit	Recommended	The packaging format (e.g., liter, gallon, sack, drum). Helps with physical stock verification.
Packaging Quantity	Recommended	Units per package (e.g., 20 liters per drum). Used to convert between purchase units and storage units.
Supplier	Recommended	Name of the supplier or distributor. Used for procurement analysis.
Document Number	Recommended	Invoice or purchase order number from the supplier. Always record this for audit purposes.
Notes	Optional	Any relevant notes about the delivery condition, lot number, or expiration date.

Link purchases to accounting

For purchases above RD$5,000, also create a corresponding transaction in Accounting > Transactions (see Chapter 4). The inventory purchase records the physical stock entry; the accounting transaction records the financial expenditure. The two records are not automatically linked — you must create both.

5.6.3 Inventory Items Reference

Current active inventory items by category:

Post-emergent herbicides

- Agromina 72 SL
- Ametrex 50 SC
- Cañarex 65 WG
- Terbutrizell
- Glifosato
- Crezendo 45.5 CS
- Rainbomina
- Revolver

- Profitol 75 WG
- Picloran
- MSMA
- Dinamic (powder)
- Dinamix (powder)
- Dinamic

Pre-emergent herbicides

- Pledge 51 WG (kg)
- Enfoke 75 WG (gram)
- Karmex
- Mayoral 35 SL

Fertilizers

- DAP (kg)
- Cover Fertilizer (kg)
- Nutrihojas (liter)

Adjuvants and conditioners

- Bivert (adherente)
- Bionex (adherente)
- Abland (condicionador)
- pH Ned (condicionador)

Fuel (managed automatically from tank levels)

- Diesel Agrícola (gallons)
- Diesel Industrial (gallons)

Duplicate item: Dinamic

There are currently two items named 'Dinamic' in the system (both post-emergent herbicides, both in kg). One has stock of 1 kg and the other 0 kg. Before recording usage of Dinamic, verify which item ID your operation should reference. Ask your administrator to merge or rename the duplicate.

5.7 Day Labor (Jornaleros)

The Day Labor module tracks work performed by temporary day workers (jornaleros) who are paid by the day or task rather than on the regular payroll. Entries are grouped by week and can be closed once verified and paid.

5.7.1 Recording a Day Labor Entry

Navigate to Operations > Day Labor > + New Entry.

Field	Required?	Description
Work Date	Required	The actual date the work was performed.
Week Ending Date	Required	The Saturday that ends the work week this entry belongs to. Used for weekly reporting and payment grouping.
Operation Description	Required	A brief description of the task performed (e.g., Chapeo manual, Siembra manual, Deshierbo). This is free text — it does not link to the formal operation types.
Field Name	Recommended	The field where the work was done. Free text (not a dropdown). Enter the field code (e.g., Java 2, BN5) for consistency.
Worker Name	Recommended	The name of the day worker. Leave blank only if recording a group total.
Workers Count	Required	Number of workers who performed this entry's task on this date. Defaults to 1.
Amount	Required	Total payment for this entry in DOP. For a group, this is the total for all workers combined.

5.7.2 Closing a Week

Once all day labor entries for a week have been verified and paid, mark the week as closed. Navigate to Operations > Day Labor, filter by the week ending date, and click Close Week.

- Closed entries are read-only and cannot be edited.
- A week can only be closed if all entries in that week are marked as verified.
- Closing a week does not automatically create an accounting transaction — you must create the payment entry manually in Accounting > Transactions.

Jornalero registry

Registered day workers are maintained in a separate Jornaleros list (Operations > Jornaleros), which records their cédula (national ID) and active status. You can record day labor entries for unregistered workers using free text in the Worker Name field, but registered workers are preferred for payroll reporting and identity verification.

5.8 Cronograma (Weekly Work Schedule)

The Cronograma is the weekly work plan for all permanent employees and registered day workers. It shows each worker's assigned task for each day and time slot (morning / afternoon) for the current week. It is the primary planning and attendance reference tool for supervisors.

5.8.1 Navigating the Cronograma

Navigate to Operations > Cronograma. The view shows the current week (Monday through Saturday) with each worker as a row and each day-and-slot as a column. Use the week arrows to navigate to previous or future weeks.

- Week ending dates run Saturday-to-Saturday. Eight weeks are currently open.
- Employees appear with type: Employee; day workers with type: Jornalero.
- Morning and afternoon slots are tracked separately.
- Days 1–6 correspond to Monday through Saturday.

5.8.2 Adding a Cronograma Entry

Click on any empty cell in the grid (worker × day × slot) to create a new entry. A small form appears:

Field	Required?	Description
Task	Recommended	The operation or task the worker is assigned to. Free text. Can be linked to a formal operation record via the source_operation_id field (advanced).
Is Holiday	Optional	Check if this slot falls on a public holiday. Affects payroll calculation.
Is Vacation	Optional	Check if the worker is on approved vacation. Affects payroll calculation.

Cronograma and payroll

The cronograma feeds directly into the payroll module. Marking a slot as Is Holiday or Is Vacation here is what triggers the corresponding pay rule (holiday premium or vacation deduction) when payroll is processed. Always keep the cronograma up to date before closing a payroll period.

5.9 Service Contracts

The Service Contracts module tracks third-party equipment or labor contracts billed by the unit — typically by hour, hectare, or ton. A contract defines the rate; individual work entries record units completed; payments link to the accounting module.

5.9.1 Creating a Service Contract

Navigate to Operations > Service Contracts > + New Contract.

Field	Required?	Description
Contract Name	Required	A descriptive name identifying this engagement (e.g., 'Canal Excavation — February 2026').
Owner / Contractor	Required	The name of the third-party company or individual providing the service.
Owner Cédula/RNC	Recommended	The contractor's tax ID. Required for DGII reporting when payments exceed reporting thresholds.
Operation Type	Required	The type of work being contracted (Excavator, Mowing, Harvest, Transport, Other).
Unit Type	Required	The billing unit: hours, hectares, or tons.
Price per Unit	Required	The agreed rate in DOP per unit.
Farm	Optional	The farm this contract is associated with.
Bank / Account	Optional	Contractor's bank details for payment reference.
Comments	Optional	Contract terms, notes, or references.

5.9.2 Recording Work Entries

Each day or session of contracted work is logged as an entry against the contract. Open the contract and click + Add Entry.

Field	Required?	Description
Entry Date	Required	Date the work was performed.
Description	Required	Brief description of work done that day.
Units Charged	Required	Number of hours, hectares, or tons completed in this session.
Calculated Cost	Auto	Filled automatically as Units Charged × Price per Unit.
Cost Override	Optional	Enter a different amount if the session cost deviates from the standard rate (e.g., emergency rate, partial day).
Comments	Optional	Any notes about the session.

5.9.3 Recording Payments

Payments to the contractor are recorded under the contract's Payments tab. Click + Add Payment.

7. Enter the payment date, amount in DOP, and any notes.
8. In the Transaction field, link the payment to an existing accounting transaction (created in Accounting > Transactions). This creates the formal audit trail between the operational record and the financial ledger.
9. Click Save. The contract's outstanding balance updates automatically.

Contract balance

The contract detail view shows total work value (sum of all entries), total paid (sum of all payments), and outstanding balance. Use this to determine when the next payment is due and to verify that all work sessions have been invoiced.

5.10 Rainfall Records

The Rainfall module captures daily precipitation readings from four weather measurement stations across the farm. This data is used for irrigation planning, field scheduling, and agrochemical application windows.

5.10.1 Recording a Rainfall Reading

Navigate to Operations > Rainfall > + New Reading. Each record covers one calendar date.

Field	Required?	Description
Date	Required	The measurement date. Each date can only have one record — attempting to save a duplicate date will return an error.
Palmarito (mm)	Recommended	Rainfall in millimeters at the Palmarito station.
Solar (mm)	Recommended	Rainfall in millimeters at the Solar station.
Virgencita (mm)	Recommended	Rainfall in millimeters at the La Virgencita station.
Caoba (mm)	Recommended	Rainfall in millimeters at the Caoba station.

Leave a station field blank (null) if that gauge was not read that day — do not enter 0 unless rain was actually measured as zero. A blank value means 'not recorded'; a zero value means 'no rain recorded'.

Rainfall and operation planning

A best practice is to enter rainfall readings every morning before logging field operations. High rainfall events (above 20 mm) typically affect whether spraying or soil-contact operations should proceed. Supervisors can reference the last 7 days of rainfall from the Cronograma view to inform the day's work plan.

5.11 Common Scenarios

Scenario	How to record it
Spraying operation with product mix	Create a Field Operation (type: Pulverizar), select the tractor and sprayer implement, enter start/end hours. After saving, open the operation and add each product as an Input (e.g., Revolver 6.2 L + Abland 0.91 L). Enter hectares done.
Manual weeding with day workers	Create a Field Operation (type: Deshierbo manual or Chapeo manual), leave Tractor blank, enter Workers Count. Also create a Day Labor entry for the same date and field to record the payment amount. The two records are separate — link them via Comments if needed.
Tractor fuel-up at the Mobile tank	Create a Fuel Transaction (type: Dispense), select Mobile tank, select the tractor, enter pump start and end readings, enter gallons, and enter the current hour meter reading. If a Driver submitted this via the portal, approve it from Pending Submissions instead.
Diesel delivery to Main Tank	Create a Fuel Transaction (type: Refill), select Main Tank, enter gallons received and supplier name in Notes. Then create an Accounting Transaction for the cost (Chapter 4), referencing the same delivery.
Oil change on JD3006	Navigate to Equipment > JD3006 > Maintenance tab > + Log Maintenance. Set type to Routine, enter today's hour meter reading, and note the oil grade and filter used. The maintenance countdown resets from this hour meter value.
Weekly jornalero payment	Verify all day labor entries for the week, then click Close Week. Create a transaction in Accounting > Transactions (pay method: Bank Transfer or Cash, account: Day Labor / Jornaleros) for the total weekly amount.
Third-party excavator hired for one day	Create a Service Contract if one does not exist, or add an entry to the existing contract. Record units (hours worked). After payment, add a Payment record linked to the corresponding accounting transaction.
Herbicide running low	Check Operations > Inventory > item detail. Note current_quantity. Create a procurement request outside the system. When the purchase arrives, record an Inventory Purchase entry. Also create an accounting Transaction for the cost.

5.12 Quick Reference — Who Does What

Task	Admin	Mgmt	Supervisor	Accountant
Create / edit field operations	✓	✓	✓	✓
Log fuel dispense / refill	✓	✓	✓	✓
Log maintenance event	✓	✓	✓	✓
Record inventory purchase	✓	✓	✓	✓
Add inventory inputs to operation	✓	✓	✓	✓
Create day labor entry	✓	✓	✓	✓
Close a day labor week	✓	✓	✓	✓
Edit cronograma entries	✓	✓	✓	✓
Create / manage service contracts	✓	✓	—	✓
Add service contract payment	✓	✓	—	✓
Record rainfall	✓	✓	✓	✓
Add / deactivate farms & fields	✓	✓	—	—
Add / deactivate equipment	✓	✓	—	—
Add operation types	✓	—	—	—
Approve driver portal submissions	✓	✓	✓	✓

5.13 Related Chapters

- Chapter 4: Registering Transactions — recording financial costs linked to operations
- Chapter 6: Journal Entries — how operational costs post to the general ledger
- Chapter 9: HR and Payroll — permanent employee time tracking and payroll processing
- Chapter 10: DGII Reports — how service contract and day labor payments appear in 606 filings